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New Cretaceous paleomagnetic poles from the Tarim Basin, Northwestern China

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ABSTRACT

Most paleomagnetic studies of the Tarim have been focused on the Paleozoic, with little data published for Mesozoic rocks. However, the ancient configuration of this part of Asia has been affected by the Indian–Eurasian collision. During the second French–Chinese Karakorum expedition, from Kashkar to Kunjerab, in the Autumn of 1990, we sampled about 500 cores from 53 sites of Upper and Lower Cretaceous red sandstones from three areas near Uytak, Yingjisha and Wuqia (38.5–39.5°N, 75–76.5°E). Isothermal remanent magnetization acquisition indicates the presence of minerals with high magnetic coercivity. Stepwise thermal demagnetization confirms that hematite is the dominant carrier and allowed the separation of low temperature components (LTC, < 300°C) from high temperature components (HTC, > 600°C) in most samples. The mean direction of LTCs is aligned with that of the present Earth field. Normal and reversed polarities have been found for the HTCs of each age group. HTCs show positive fold tests for at least one sub-study where the attitudes of bedding were sufficiently different to allow the test to be performed. More important, the four sub-studies from Yingjisha and Wuqia give fully compatible results. Results from the Uytak area show a wide dispersion in declinations, as could be expected from the rather severe deformation and proximity to major thrust and strike-slip faults. Data obtained in the Kuche area from northern Tarim by Li and co-workers and in southern Junggar by Chen and co-workers are statistically identical to the new data from Yingjisha and Wuqia. Data from Paicheng in northern Tarim appear to have been subject to local rotation. Excluding those sites which appear from paleomagnetic study to have been significantly rotated, we calculated a mean Cretaceous pole for the Tarim basin at 69.1°N, 223.8°E ($A_{95} = 3.5$°, $N = 5$). This result indicates that the total post-Cretaceous rotation of Tarim relative to Junggar and to Eurasia has been negligible (2.6 ± 6.1° and 0.4 ± 4.6°, respectively). Continental shortening and rotation between Tarim and Junggar, within the Tien Shan, are not resolved by paleomagnetism although the values obtained (400 ± 600 km, 3 ± 6°) are compatible with tectonic models of the range. A major discrepancy, warranting further geological investigation, is that between the Tarim and North China block poles, which would imply some 1400 ± 1060 km of north–south relative displacement.

1. Introduction

The Tarim basin (Fig. 1), the largest basin in China, has long remained essentially out of bounds for geological and geophysical work, largely because it is a remote cold desert area with very harsh natural conditions. Geological exploration began in the 1950s and geophysical work had to wait until the 1980s. Paleogeographic and tectonic comparisons have led Chinese authors to propose various interpretations [3,4]. Li [4] is one of many to have proposed that the Tarim belonged to a single Sino–Korea–Tarim plate since the late Paleozoic. Early paleomagnetic work soon contradicted this view: the first Permian results from the Tarim [5] were significantly different from those of McElhinny [6] for the North China block (NCB). This has subsequently been supported by further results [7–10]. Most of this work has been concerned with the Paleozoic assemblage of microcontinental blocks; it is, however, obvious from the local tectonics

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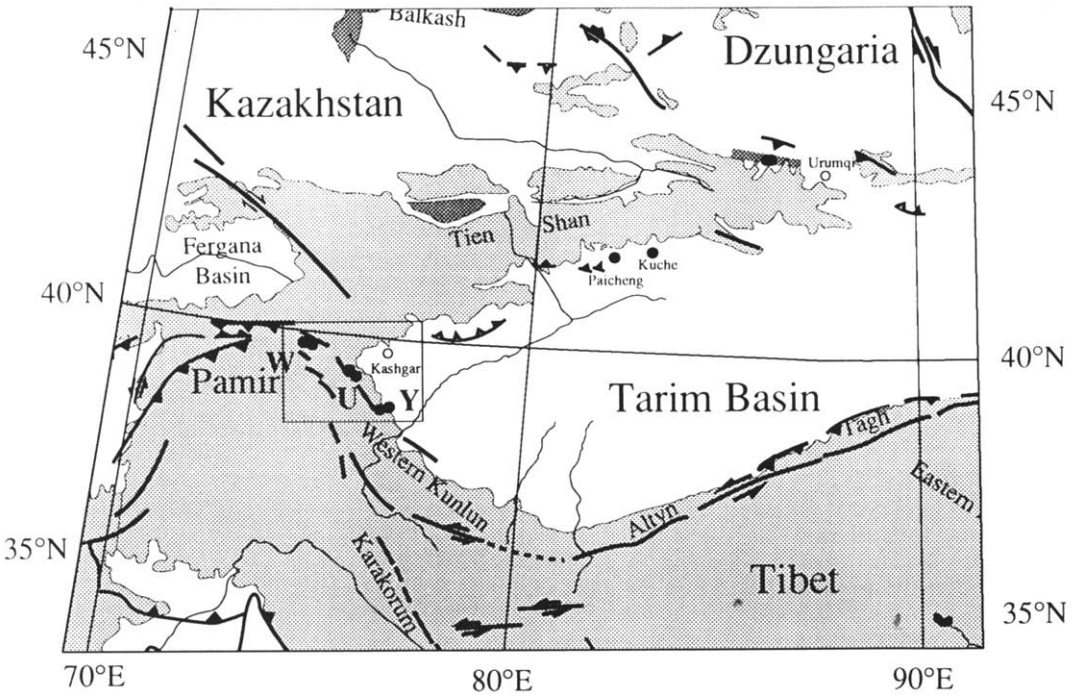


Fig. 1. Schematic tectonic and topographic map of the Tarim basin and its surroundings, after [13]. Dots = sampling localities; *U*, *W* and *Y* = Uytak, Wuqia and Yingjisha, respectively.

that the ancient configuration of the mosaic must have been severely affected by the collision of India with Eurasia [e.g., 11–13]. Older faults and sutures were reactivated, while thrusting, folding and mountain build-up, and plateau formation took place. There has been a general lack of paleomagnetic observations for the Mesozoic and Cenozoic periods which could quantitatively constrain this more recent tectonic deformation; with the exception of Li et al. [1] in the northern Tarim and Chen et al. [2] in the southern Junggar basin, just north of Tarim. During the second French–Chinese Karakorum–Kunlun expedition from Kashgar to Kunjerab, in the Autumn of 1990, we attempted to extend the data base to westernmost Tarim (Fig. 2).

2. Geology and sampling

The Tarim basin, which extends over 560,000 km^2, has a rhombic shape, framed by the Tien Shan, Kunlun and Altyn Tagh ranges (Fig. 1). A late Paleozoic (Hercynian) orogeny was responsible for a general unconformity between Paleozoic and Mesozoic formations: the Tarim has been an inland basin since the beginning of the Mesozoic. It was profoundly modified as an indirect result of the Asia–India collision. Southward subduction of the Tarim below the Kunlun created a foredeep basin with more than 7000 m of Tertiary–Quaternary sediments. Intracontinental deformation was also responsible for extensive faulting and folding and disconformities, as recently as the Pliocene and early Quaternary. The collision, which may have begun as early as the earliest Cenozoic [e.g., 14], is clearly responsible for the tectonic deformation of the Mesozoic strata, which were the main focus of the present study.

In the general sampling area (Fig. 2), a continuous series of Cretaceous outcrops is widely exposed. This series consists mainly of red and dark red sandstones, interbedded occasionally with variegated mudstones, gypsum shales and limestones. The lower Cretaceous corresponds to the Kezilesu Formation, separated from the upper Cretaceous Yingjisha Formation by the characteristic Uytak Group. Dating is based on paleonto-

logical research carried out by the Nanjing Institute of Paleontology and the Xinjiang Bureau of Geology and Resources [15; D.L. Sun, personal communication, 1990]. Samples of both upper and lower Cretaceous were collected from three widely separated areas (Fig. 2): Uytak area

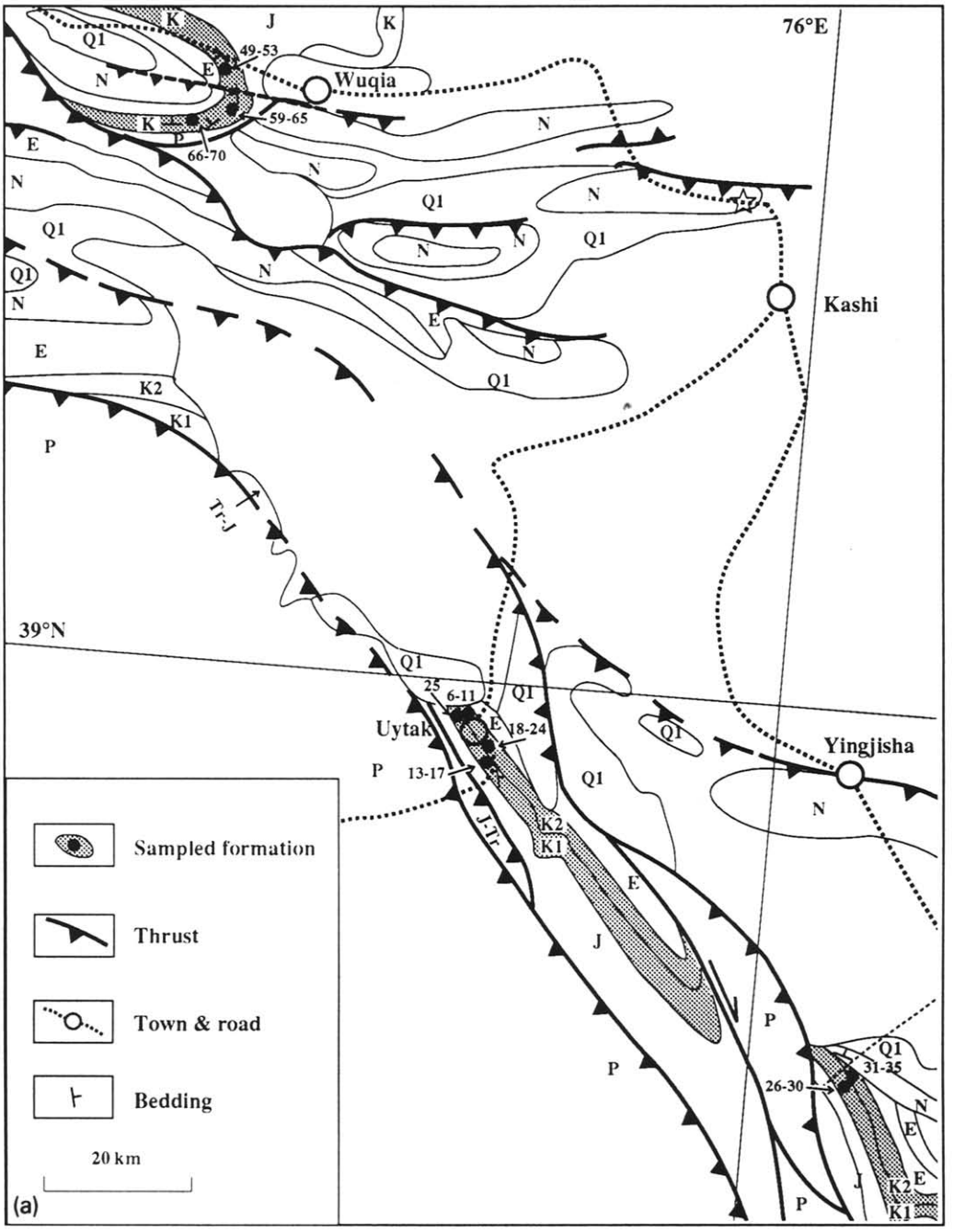
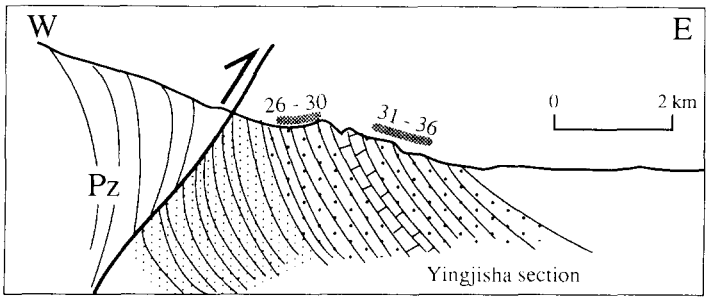
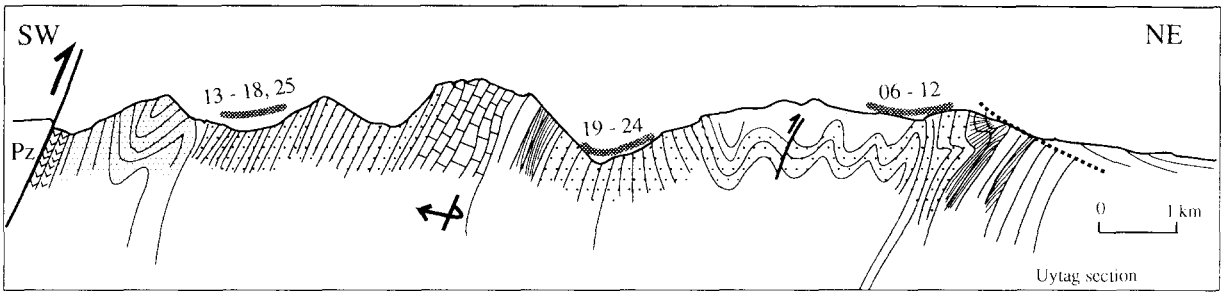
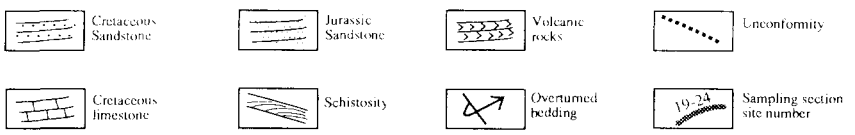


Fig. 2. (a) Simplified geological map of sampling area. Cretaceous formations are shadowed and site numbers are indicated. Formation ages: *Q1* = lower Quaternary; *N* = Neogene; *E* = Paleogene; *K* = Cretaceous (*2* = upper, *1* = lower); *J* = Jurassic; *Tr* = Triassic and *P* = Paleozoic or older; star north of Kashi = site location in Pliocene Atushi formation.

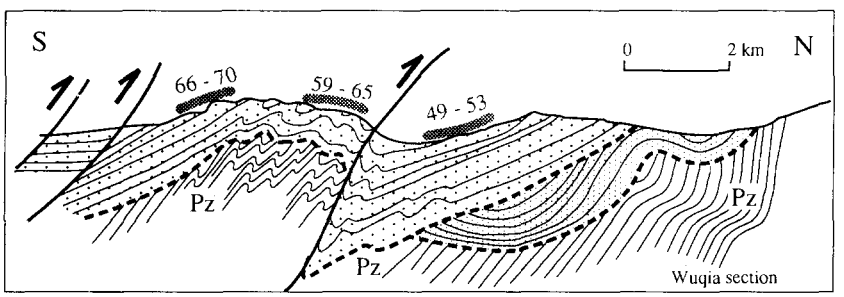


(b)





(c)



(d)

Fig. 2. (b) Cross-section of sampled area near Yingjisha. (c) Cross-section of sampled area near Uytak. (d) Cross-section of sampled area near Wuqia [36].

(38.9°N, 75.5°E), with 13 upper (Ku) and 7 lower Cretaceous (Kl) sites; south of the Yingjisha area (38.5°N, 76.4°E), with 6 Ku and 5 Kl sites respectively; and the Wuqia area (39.6°N, 75.0°E), with 15 Ku and 7 Kl sites. For two of these sub-studies, fold limbs were sampled so that a fold test could be attempted (Upper Cretaceous from the Uytak and Wuqia areas).

Dating is reasonably well constrained in some areas, such as Yingjisha, where, for instance, dinosaur prints have been found. In others fossils were not found and we had to rely on local

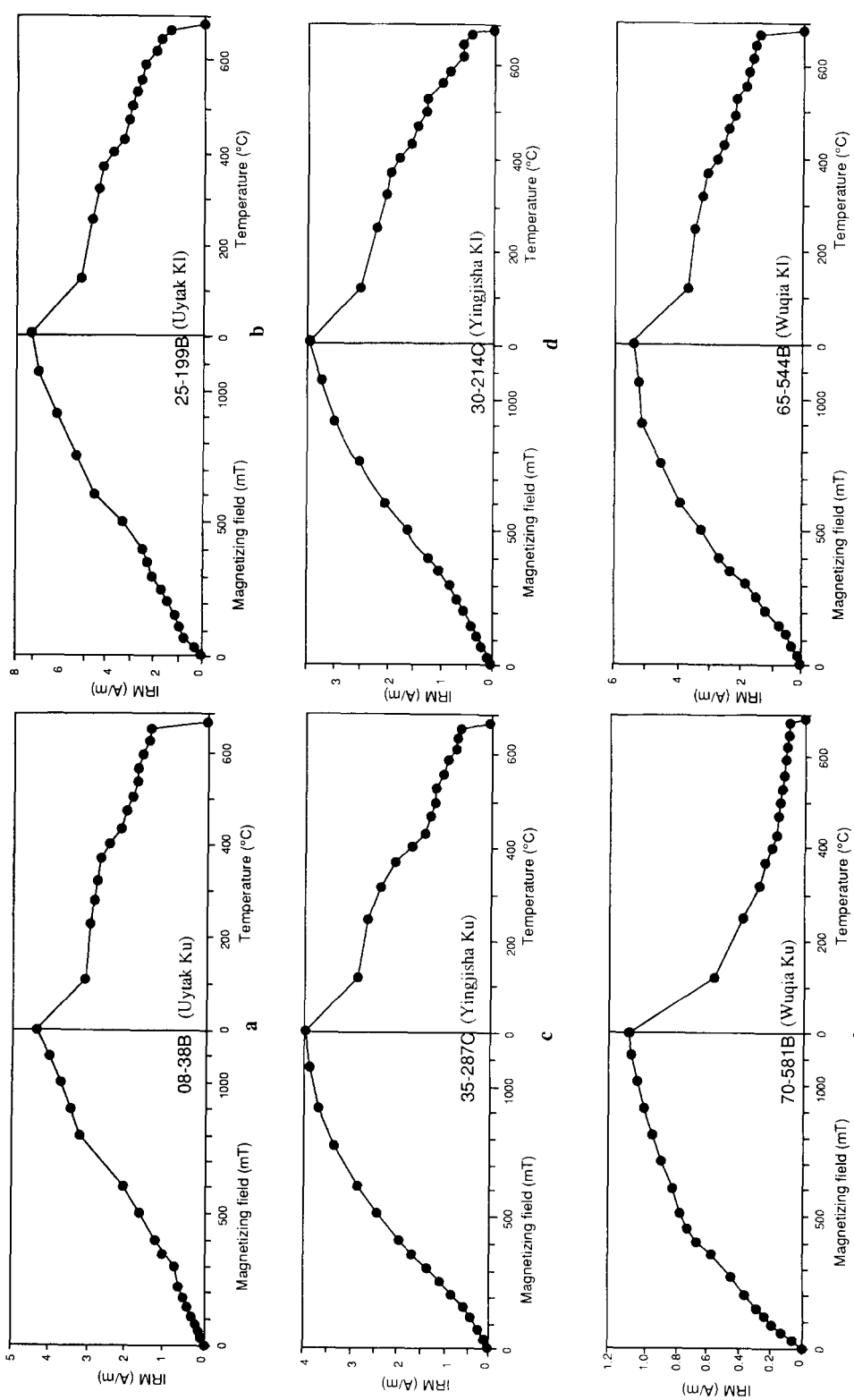


Fig. 3. Acquisition and thermal demagnetization of isothermal remanent magnetization (IRM) showing similarity of magnetic mineralogy in the three localities sampled. (a) and (b) Uytak. (c) and (d) Yingjisha. (e) and (f) Wuqia. For two stratigraphic age groups (a), (c) and (e) are Ku, (b) (d) and (f) are Kl.

mapping and facies similarities. As a result, we cannot be sure that some of our "lower" Cretaceous sites do not extend into the upper Jurassic and that some of our "upper" Cretaceous sites do not extend into the Eocene.

The structural setting of our sites is derived from published maps [16] and from further field work [12] (Fig. 2a). The southernmost sites south of Yingjisha are part of a fairly simple anticlinal structure (Figs. 2a and b); all were collected on a limb dipping some 30° towards the Tarim Basin. Except for minor possible rotation about a vertical axis, these sites are thought to belong to the Tarim block. They are separated from the Uytak sites by a major thrust and right-lateral strike-slip fault system (Fig. 2a). The Uytak sites belong to a strongly folded structure; more precisely, the overturned flank of a large-scale, plunging syncline. Dips vary from 30° to 130° and thirteen out of nineteen sites were in a overturned position (Fig. 2c). This more complex geometry of deformation will have to be kept in mind when analysing the paleomagnetic data. The Wuqia sites were collected from a fairly simple east–west trending synclinal structure, with a rather broad range of strikes and dips (Fig. 2d). The folding between the Pamir and Tien Shan is quite significant, although more straightforward than at Uytak. A priori, these sites are expected to have suffered less intense deformation and displacement relative to the Tarim than those at Uytak.

In addition to these Cretaceous sites, five sites of muddy sandstones were collected from the Atushi Pliocene Formation north of Kashgar (or Kashi; 39.5°N, 76.0°E) and eight sites from slightly metamorphic Permian volcanic clastics, about 60 km south of Taskurgan town (37.2°N, 75.5°E). These last two sub-studies provided little or no information and are excluded from this report (for more details see [17]).

In each site, 2.5 cm diameter cores, 5–10 cm long, were collected with a gas-powered drill. Usually, eight to ten cores were drilled at each site, spanning over 10 m in stratigraphic thickness. Cores were oriented with a magnetic compass and a solar azimuth was measured whenever possible. The differences between both determinations of azimuths for 310 samples average 3.5°E, with a variance of 1.2°, which can be compared to the declination calculated from the 1985 IGRF (corrected to August 1989) at a reference of 39°N, 75.5°E, which is 1.0° (the inclination being 55.6°). This average was used to correct all measurements without a solar azimuth, in particular bedding strikes.

3. Paleomagnetic techniques

Cores were cut into 2.5 cm long specimens that were measured conventionally. Since hematite was the dominant magnetic mineral, alternating field (AF) demagnetization almost always proved to be inefficient and stepwise thermal demagnetization was preferred to isolate magnetic components. A Schönstedt AF demagnetizer and two nearly zero-field laboratory built furnaces were used for these experiments. Magnetic measurements were performed with a CTF 3-axis cryogenic magnetometer. Bulk susceptibility was measured after each heating step to monitor possible changes in magnetic minerals; most of the time no significant change occurred over the 700°C range under study. In order to get additional constraints on magnetic mineralogy, IRM acquisition experiments were carried out and subsequent thermal destruction of the acquired IRM was observed [e.g., 18]. Magnetic directions were extracted by principal component analysis [19] and remagnetization circles [20]. These data were incorporated to calculate site-mean directions using the statistics of Fisher [21] and the method of McFadden and McElhinny [22]. Data acquisition and processing were performed using the computer software package of Enkin [23] developed in our laboratory.

4. Paleomagnetic results

The Lower and Upper Cretaceous samples from the Uytak, Yingjisha and Wuqia areas respond in a remarkably similar manner to the magnetic and heating experiments. Figure 3 brings together the IRM acquisition and thermal decay experiments. Despite some second-order differences, IRM acquisition curves have a S shape with an inflexion point near 500–600 mT. In most cases, except perhaps the Lower Cretaceous of Wuqia, there is no indication that saturation has been reached at 1–1.2 T. High coercivity minerals therefore dominate the magnetic mineralogy, al-

though a lower coercivity phase may be present in the Lower Cretaceous of Uytak and in the Upper Cretaceous of Wuqia (which has the most different behaviour compared to the other sites).

The thermal decay of the IRM has the same general shape, with a fast decay from room temperature to 120–150°C, an inflexion near 400°C and a sharp drop to zero at 660–680°C. Only the

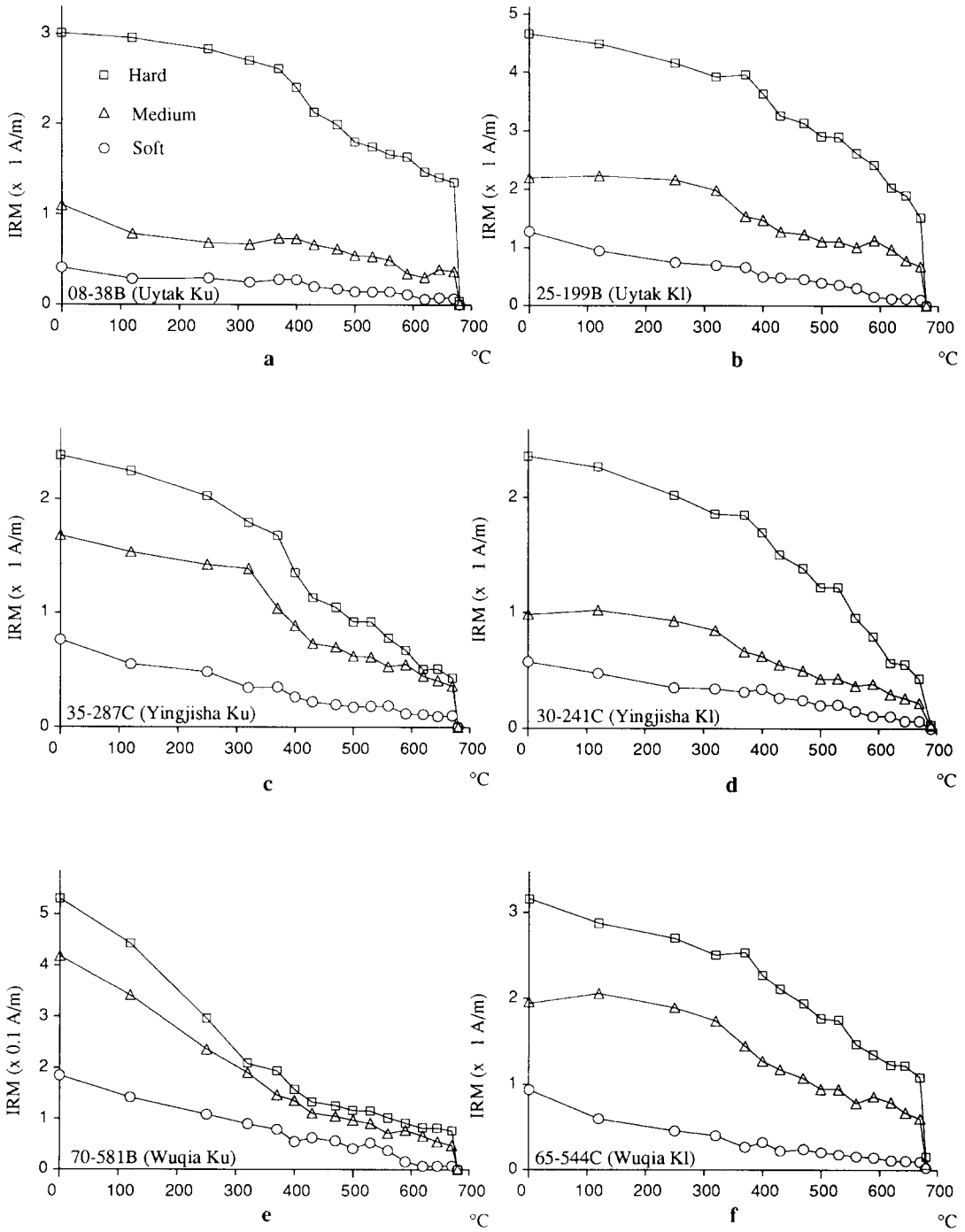


Fig. 4. Three-component IRM thermal demagnetization [24], indicating two main magnetic intensity drops near 400°C and 680°C.

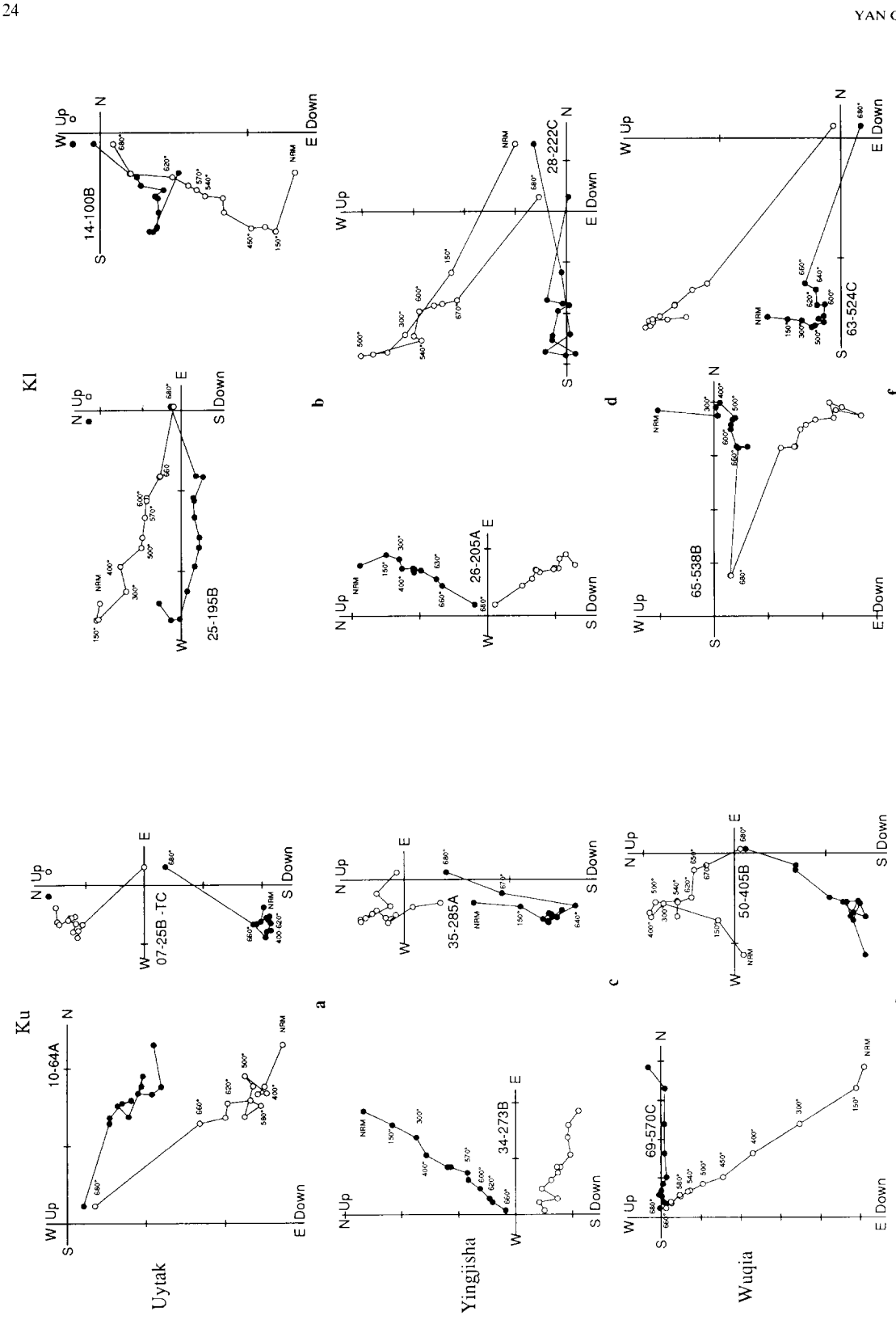


Fig. 5. Orthogonal vector projections of representative thermal demagnetizations, displaying the presence of high unblocking-temperature minerals. All diagrams are in stratigraphic coordinates with scale bars 0.001 A/m apart.

relative amplitudes of the three drops (at 120, 400 and 680°C) differ. After removal of a component with low unblocking temperature and high coercivity, which was most probably viscous and/or carried by goethite, the spectrum is dominated by hematite. The component which becomes unblocked near 400°C is minor and may indicate that some sulphides, titanomagnetites or larger grain size hematite are also present.

Thermal demagnetization of multi-component IRM, imparted following the method of Lowrie [24], confirms these observations (Fig. 4). A "hard" component disappears about 400°C, particularly in the Uytak and Yingshija Ku and Kl samples. In the latter case, the same unblocking temperature (T_{ub}) range is found in the "medium" component. The Wuqia Ku sample has large "hard" and "medium" components with T_{ub} less than 300°C, apparently corresponding to low coercivities. There is very faint evidence for the presence of magnetite in the "soft" component of the Uytak Kl, Yingshija Ku and, maybe also, Wuqia Ku samples.

Although some demagnetization diagrams show erratic behaviour and reveal remagnetization in the laboratory upon thermal treatment, most specimens are quite well behaved and reveal

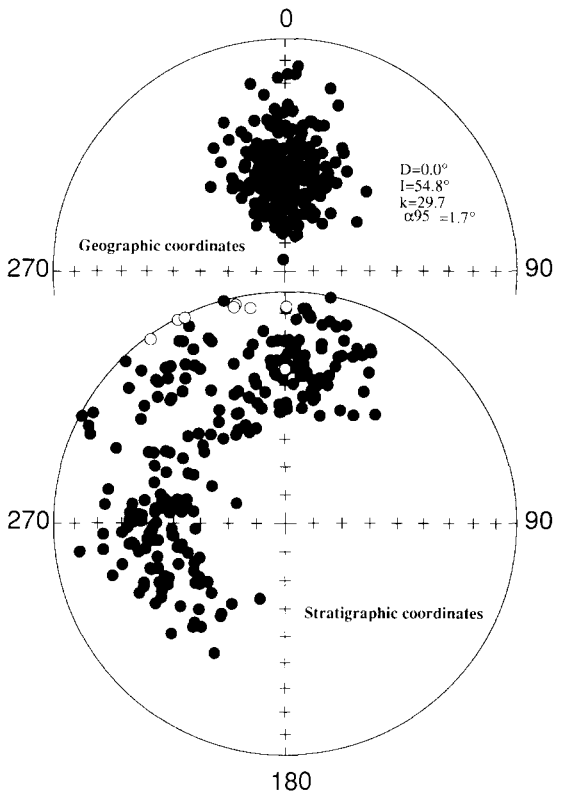


Fig. 6. Equal-area projection of 245 sample directions (low-temperature component) in both geographic and stratigraphic coordinates.

TABLE 1

Mean paleomagnetic directions for the lower temperature component

Area	Age	n	Dg	Ig	Ds	Is	k	α_{95}	kg/ks
Uytak	Ku	62	357.4	49.9	–	–	28.8	3.4	
Uytak	Ku	62	–	–	312.2	27.6	8.6	6.5	3.38
Uytak	Kl	30	2.3	62.7	–	–	31.7	4.7	
Uytak	Kl	30	–	–	255.7	38.3	17.4	6.5	1.82
Yingjisha	Ku	39	0.1	52.6	–	–	51.5	3.2	
Yingjisha	Ku	39	–	–	12.2	28.9	36.6	3.8	1.41
Yingjisha	Kl	22	357.3	56.6	–	–	35.0	5.3	
Yingjisha	Kl	22	–	–	5.1	30.9	31.5	5.6	1.11
Wuqia	Ku	58	359.6	57.5	–	–	27.4	3.6	
Wuqia	Ku	58	–	–	326.1	52.3	3.6	11.6	7.61
Wuqia	Kl	34	5.4	53.6	–	–	30.9	4.5	
Wuqia	Kl	34	–	–	275.9	47.6	27.8	4.8	1.11
Overall average									
Samples	K	245	0.0	54.8	–	–	29.7	1.7	
Samples	K		–	–	320.0	44.4	4.1	5.1	7.24
Localities	K	6	0.4	55.5	–	–	285.2	4.0	
Localities	K		–	–	321.1	46.3	5.0	33.4	57.04

n = number of specimens; Dg, Ig = declination and inclination in geographic coordinates; Ds, Is = same in stratigraphic coordinates; k, α_{95} = Fisher precision parameter and radius of the 95% confidence; kg/ks = ratio of precision parameter in geographic and stratigraphic coordinates.

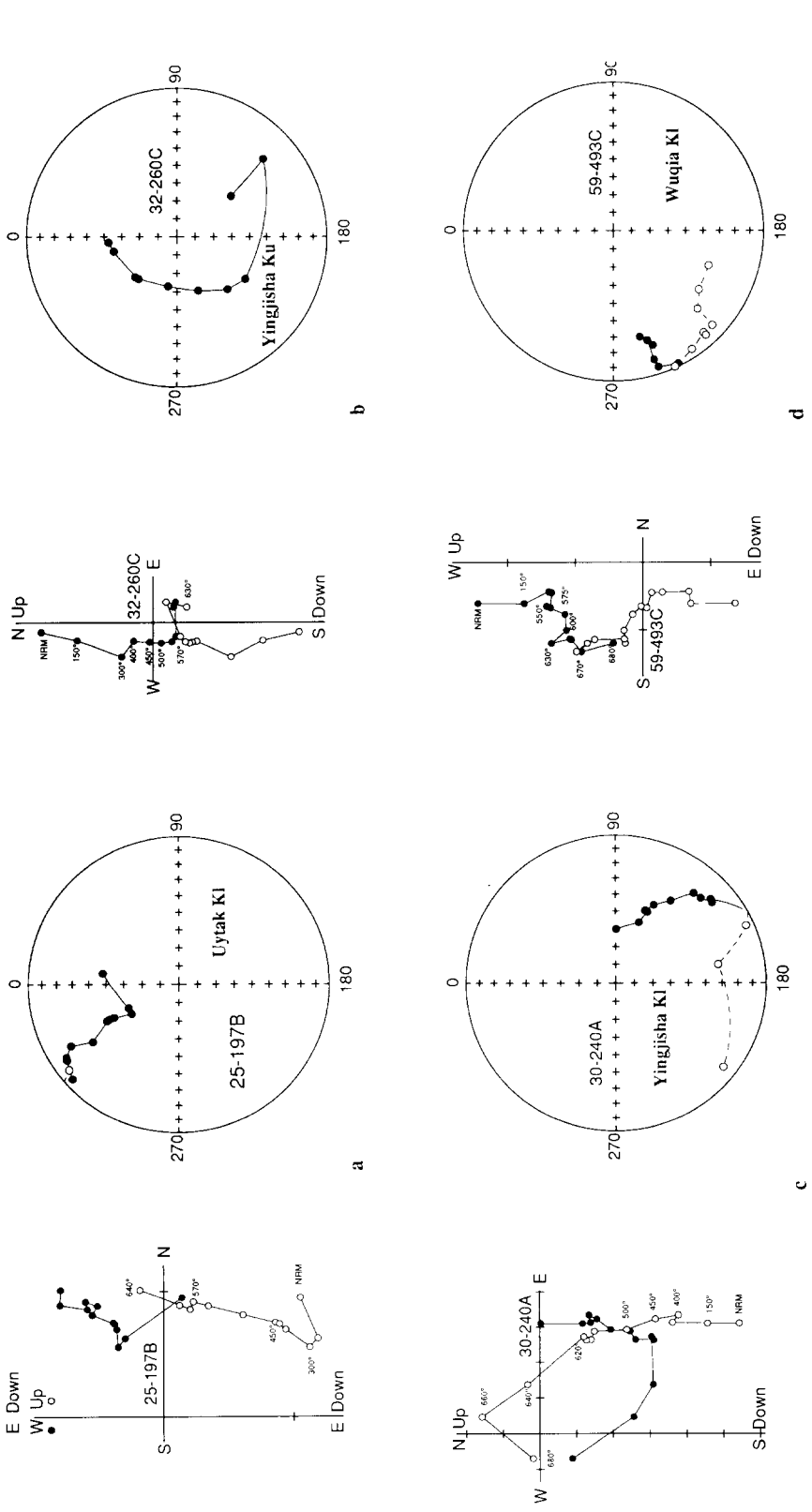


Fig. 7. Representative orthogonal vector projections and equal-area diagrams illustrating cases when the high temperature component could not be successfully isolated by principal component analysis but where the remagnetization circles technique could be used. All diagrams are in stratigraphic coordinates, with scale bars 0.001 A/m apart.

TABLE 2

Mean paleomagnetic directions for Upper Cretaceous sites from the Uytak area

Site	n/N	P	Dg	Ig	s	d	Ds	Is	k	α_{95}	ks/kg
06 *	9/9	R	37.8	−33.5	356	100	310.2	−26.8	6.2	22.8	
07 *	5/6	R	125.6	3.2	356	121	203.2	−42.8	8.1	32.1	
08 *	9/10	N	59.5	2.9	194	45	47.7	32.2	5.9	23.4	
09	7/9	R	267.1	44.7	185	62	272.0	−15.0	14.4	17.5	
10	8/8	N	59.2	40.0	215	32	28.9	45.6	31.0	10.1	
11	8/8	N	47.0	35.4	214	37	20.4	35.6	28.2	10.7	
12 *	7/7	N	52.8	−1.2	196	66	32.4	34.2	7.4	23.9	
19 *	5/6	R	21.2	−41.5	323	120	256.9	−12.7	14.4	23.5	
20	6/7	R	352.8	18.1	323	132	317.1	−34.0	18.1	15.5	
21 *	7/10	R	45.3	−31.5	330	119	254.2	−27.9	10.9	20.3	
23	4/7	R	37.4	−36.4	353	97	304.2	−29.1	27.5	18.9	
24 *	8/9	N	78.7	−11.6	184	60	72.4	45.3	7.0	22.8	
Averages											
A	12			3.5					4.2	25.4	
								32.3	27.8	8.8	6.6
B	5			15.7					2.8	68.3	
								36.3	11.2	11.2	4.0

Column headings as in Table 1, with: n/N = number of samples used/measured or number of sites in the grand mean; P = polarity N = normal, R = reversed; s, d = mean strike and dip of bedding (when d > 90°, the bedding is overturned).
* Sites in which $\alpha_{95} > 20°$.
Averages: A = all sites; B = same as A, without sites for which $\alpha_{95} > 20°$.
$\alpha 1-2 = 14.5°$.
$\alpha 1-3 = 29.2°$.
Azimuth axis = 93.7.
Bingham statistics of non-Fisher distribution is significant.

(in general) two magnetic components (Fig. 5). A low unblocking temperature component (LTC) ranging from room temperature to 150–300°C is almost always easily isolated. Individual directions for these components are always close, if not identical, with the direction of the present

TABLE 3

Mean paleomagnetic directions for Lower Cretaceous sites from the Uytak area

Site	n/N	P	Dg	Ig	s	d	Ds	Is	k	α_{95}
13	6/6	N	208.4	28.1	318	114	68.6	32.7	22.1	14.6
14	8/10	N	216.2	8.6	331	114	100.6	49.2	27.2	10.8
15	9/9	N	185.6	18.7	313	110	88.8	36.7	18.1	15.5
16 *	8/9	R	287.9	−3.0	293	95	278.5	−82.8	3.1	43.3
17	8/8	N	202.7	2.5	312	99	102.0	67.6	40.4	8.8
18	7/7	N	16.6	3.4	323	114	335.8	33.1	24.3	10.7
25	9/9	R/N	80.7	−45.0	345	106	249.5	−28.8	9.0	18.2
Averages										
A	7			13.3				12.1	20.0	
							49.5	6.8	27.7	
B	6			16.0				10.6	24.6	
							42.9	13.3	21.7	

Column headings as in Table 2.
$\alpha 1-2 = 17.6°$.
$\alpha 1-3 = 25.2°$.
Azimuth axis = 140.3.
Bingham statistics of non-Fisher distribution is significant.

earth field (PEF; $D = 1°$, $I = 53°$) or dipole field (DEF; $D = 0°$, $I = 56°$) (Table 1 and Fig. 6). The overall mean for the six sub-studies (two periods and three areas) is $D = 0.0°$, $I = 54.8°$ ($n = 245$ specimens; $\alpha_{95} = 1.7°$). This has at least the advantage of confirming that orientation errors are not significant!

A high unblocking temperature component (HTC) is often rather easily isolated between 300–500 and 680°C, although this characteristic component may appear as early as 150°C (Fig. 5b and e) or as late as 660°C (Fig. 5a and f). There is sometimes overlap between the LTC and HTC spectra, but the remagnetization circle technique can then always be used (Fig. 7).

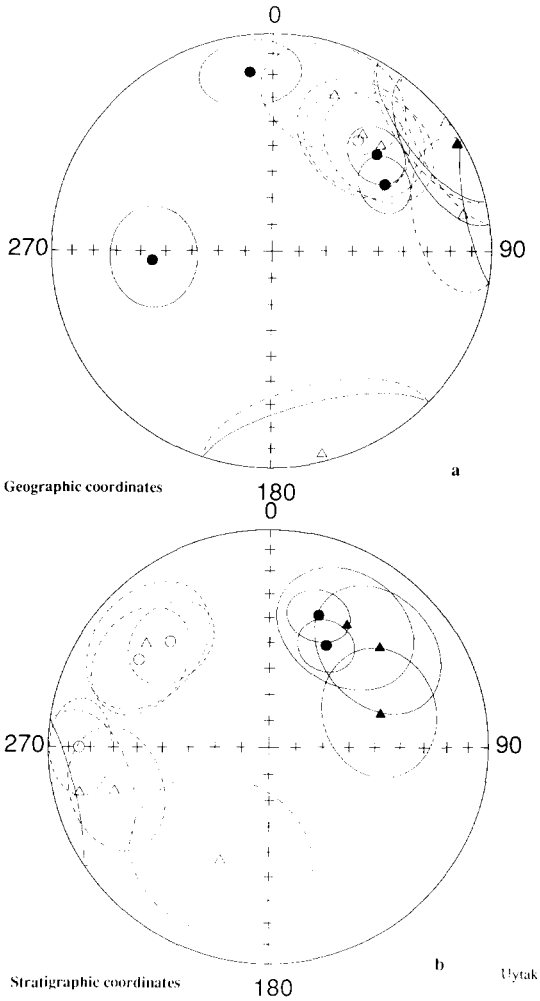


Fig. 8. Equal-area projection of site-mean directions (high-temperature component) from the Upper Cretaceous of the Uytak area (a) before and (b) after bedding corrections. Dots = site-mean directions with $\alpha_{95} < 20°$; triangles = site-mean direction with $\alpha_{95} > 20°$; black symbols and ovals of confidence = directions in the lower hemisphere; white symbols and dashed ovals = directions in the upper hemisphere.

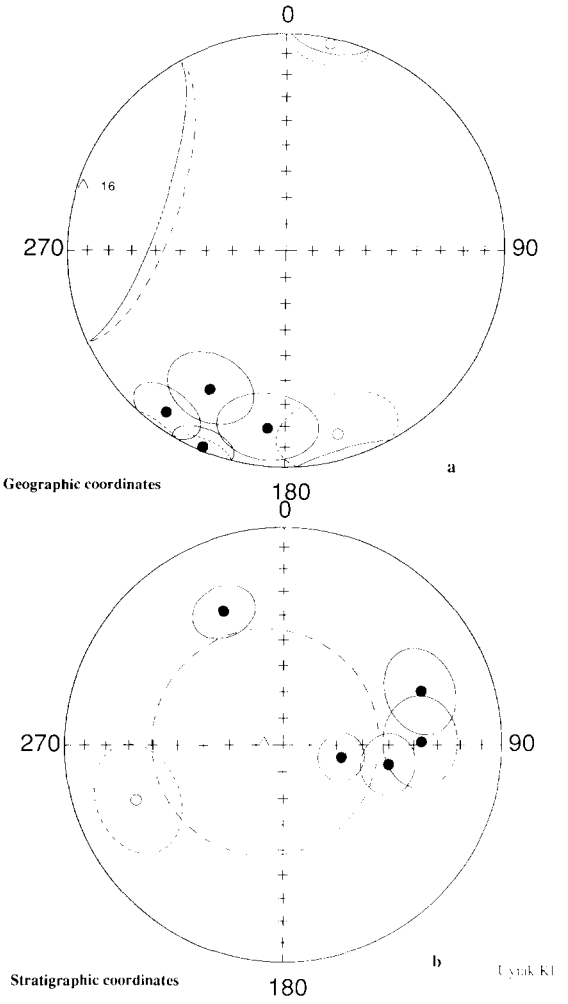


Fig. 9. As in Fig. 8 but for the Lower Cretaceous sites of the Uytak area.

Site-means are displayed in Figs. 8–13 and listed in Tables 2–7. The thirteen upper Cretaceous sites from the Uytak area are shown in Fig. 8. Unfortunately, six of them have rather large uncertainties ($\alpha_{95} > 20°$). From Fig. 8, it is obvious that tilt correction does not induce a clustering of these means. The Bingham test [25] shows that the distribution of tilt corrected directions is not Fisherian [21]. Therefore a mean Upper Cretaceous direction could not be determined for

this region. However, in stratigraphic coordinates, it appears that the dispersion mainly affects declination, whereas inclinations (reversed and normal) are similar for all sites. This is confirmed by the statistics [26] given in Table 2. We believe that this situation arises from local rotation about vertical axes, due to the rather complicated tectonic setting, as noted above.

Figure 9 shows seven Lower Cretaceous sites from the Uytak area, with a distribution similar to those of the Upper Cretaceous. The distribution of site-mean directions is obviously non-Fisherian, with a large declination dispersion. Indeed, the dispersion is so great that it does not allow us to compute an age-mean direction and only an average of inclinations is given in Table 3.

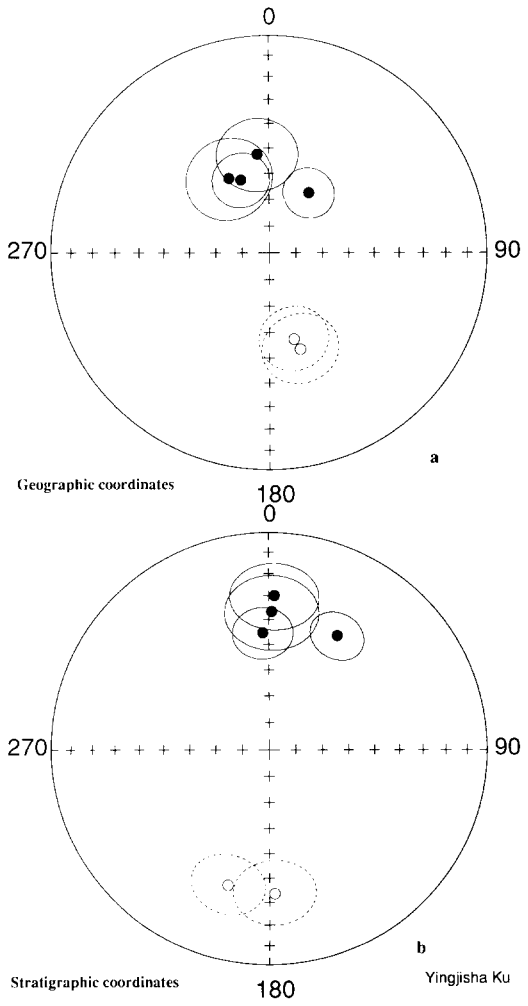


Fig. 10. As in Fig. 8 but for the Upper Cretaceous sites of the Yingjisha area.

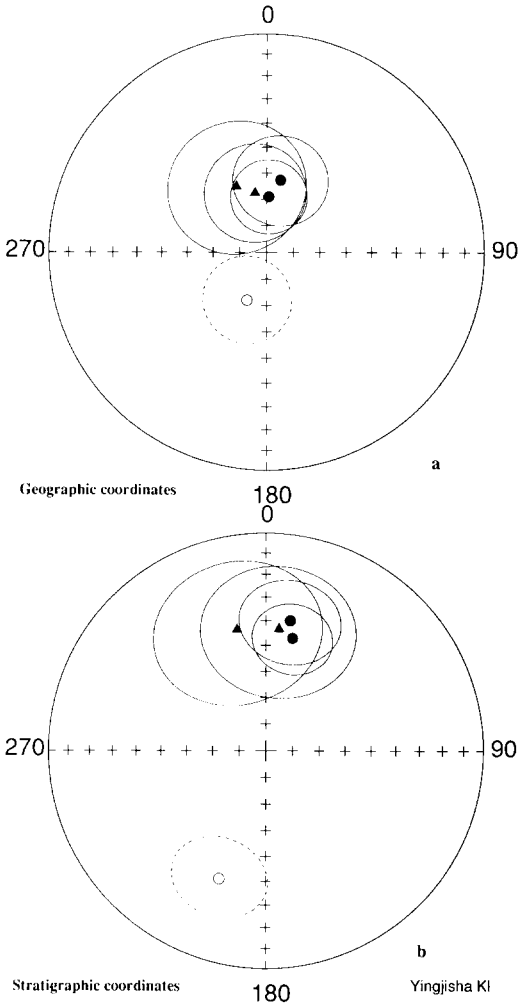


Fig. 11. As in Fig. 8 but for the Lower Cretaceous sites of the Yingjisha area.

The next two figures (Figs. 10 and 11) are for the Upper and Lower Cretaceous sites from the Yingjisha area. The six Upper Cretaceous sites are well behaved, showing both normal and reversed polarities. In one case both polarities are observed in a single site. Samples from the Lower Cretaceous were less well behaved and in two out of five sites the 95% confidence interval is larger than 20°. Due to the monoclinal setting of the sampled section, no fold test is available for these data. However, the site means form antipodal clusters, and the Upper and Lower Cretaceous

TABLE 4

Mean paleomagnetic directions for Upper Cretaceous sites from the Yingjisha area

Site	n/N	P	Dg	Ig	s	d	Ds	Is	k	α_{95}
31	7/8	N	331.8	58.1	309	33	1.2	36.7	20.0	15.8
32	6/7	N/R	339.0	60.6	299	20	357.0	45.4	69.0	10.5
33	6/6	N	33.7	62.9	298	24	31.1	39.5	50.6	9.5
34	9/9	R	164.0	−56.0	327	37	196.9	−34.5	20.4	12.5
35	8/9	N/R	162.1	−51.5	304	24	177.9	−33.5	20.2	13.7
36	8/9	N	353.3	52.3	291	24	2.1	29.8	16.7	14.5
Average										
	6		349.0	58.3			7.6	37.1	46.6	9.9

Column headings as in Table 2.

data show better consistency after the bedding correction has been applied (Tables 4 and 5).

The Wuqia area is located about 100 km West of Kashgar (Fig. 2) in the westernmost part of the Tarim Basin, in a narrow tectonic corridor between the Tien Shan and Pamir mountains, an area of intense deformation to the North of the Pamir syntaxis. Paleomagnetic results from fifteen upper Cretaceous sites are displayed in Fig. 12. Except for two sites, the results are rather straightforward (Table 6): sites are of normal polarity (except for site 50, with only two specimens) and pass a fold test at the 99% confidence level [27]. Two sites (51, 52) have a large uncertainty. One site, which has a within-site positive fold test, shows a discordant direction (52), and has been rejected from further analysis.

Data from three of the seven Lower Cretaceous sites from the Wuqia area (Fig. 13) are rather scattered (α_{95} exceeding 20°); because of significant overlap of the LTCs and HTCs and small intensity of the HTCs, the great circle constraints were not sufficient to extract the HTCs fully. The data display both polarities, but the results of the fold test are not conclusive. If we consider all site-mean directions, the fold test is positive at the 99% confidence level, but if we discard the sites with α_{95} greater than 20°, the test becomes inconclusive (Table 7). However, due to the consistency of these data with results from the Yingjisha area and with the Upper Cretaceous results from the Wuqia area itself, we consider these Lower Cretaceous results to be pre-tectonic magnetizations.

Since it was impossible to obtain a mean direction for the Uytak data, only the four mean directions obtained from the Yingjisha and Wuqia areas are shown in Fig. 14. We have calculated separately averages including sites with an α_{95} larger than 20° and excluding them (Tables 4–7). The number of these ill-determined sites ranges from zero out of six sites (for Ku of Yingjisha) to

TABLE 5

Mean paleomagnetic directions for Lower Cretaceous sites from the Yingjisha area

Site	n/N	P	Dg	Ig	s	d	Ds	Is	k	α_{95}	ks/kg
26	7/7	N	11.0	62.3	281	24	10.7	39.2	14.5	17.4	
27 *	9/9	N	335.4	62.5	274	22	346.5	42.1	10.6	29.4	
28	9/9	R	202.2	−70.6	289	33	200.9	−37.5	12.0	16.5	
29 *	5/9	N	349.1	67.1	295	26	6.0	43.1	9.4	27.0	
30	5/7	N	2.4	69.1	294	24	13.4	46.0	63.4	14.1	
Averages											
A	5		358.9	67.2			–	–	97.3	7.8	
			–	–			7.7	42.2	63.1	9.7	0.65
B	3		11.6	67.5			–	–	201.9	8.7	
			–	–			14.8	41.0	192.7	8.9	0.95

Column headings as in Table 2.

TABLE 6

Mean paleomagnetic directions for Upper Cretaceous sites from the Wuqia area

Site	n	P	Dg	Ig	s	d	Ds	Is	k	α_{95}	ks/kg
49 [1]	2/5	N	11.2	−21.1	108	76	6.9	54.1	–	–	
50 [1]	2/5	R	201.2	42.8	110	72	201.2	−29.6	–	–	
51 *	4/9	N	16.7	0.8	134	59	358.0	50.0	7.6	40.8	
52 *.[2]	8/8	N	51.5	24.2	130	49	71.0	71.8	6.6	23.3	
53	8/11	N	34.4	−22.3	122	62	34.8	39.6	15.4	15.1	
54	3/5	N	75.5	77.1	268	45	15.1	40.9	130.5	10.8	
55	5/5	N	77.4	85.4	275	50	11.0	38.5	30.3	14.1	
56	4/4	N	343.1	76.2	278	54	2.1	23.4	27.5	18.9	
57	4/4	N	4.1	81.0	278	46	9.6	34.7	35.6	15.6	
58	3/4	N	130.3	88.1	268	44	0.6	47.2	438.3	12.0	
66	6/6	N	21.1	46.1	198	30	352.8	39.8	19.8	15.4	
67	7/7	N	33.0	43.6	204	19	15.7	42.9	55.6	8.5	
68	8/8	N	25.4	49.2	217	17	8.5	43.4	10.1	19.0	
69	6/7	N	43.4	57.3	203	26	1.6	56.8	133.0	5.8	
70	8/8	N	28.8	43.4	281	17	25.4	27.1	24.3	11.6	
Averages											
A	15		28.3	43.0			–	–	3.7	23.3	
			–	–			12.3	43.5	24.2	7.9	6.5
B	14		26.9	48.6			–	–	3.5	25.8	
			–	–			11.7	40.4	38.6	6.8	11.0
C	11		30.8	60.1			–	–	6.7	19.0	
			–	–			11.0	40.0	39.5	7.4	5.9

Column headings as in Table 2.
[1] n = 2.
[2] Outlier.

four out of fifteen sites (for Ku of Wuqia). Fortunately, these averages are all identical, with angular differences of less than 3°. In addition to the sub-study for which a positive fold test is available, the individual means pass a "regional" fold or consistency test (Table 8). Because of these tests, which include the reversal test, the magnetization corresponding to the HTCs clearly pre-

TABLE 7

Mean paleomagnetic directions for Lower Cretaceous sites from the Wuqia area

Site	n/N	P	Dg	Ig	s	d	Ds	Is	k	α_{95}	ks/kg
59 *	8/8	R	110.0	−55.0	142	57	204.1	−43.0	6.1	24.7	
60	7/7	N	20.8	7.5	138	55	359.3	53.0	58.1	8.5	
61 *	4/7	R	159.0	−72.0	144	52	210.8	−30.0	12.9	28.1	
62 *	5/6	R	177.9	13.2	143	57	174.9	−19.7	9.7	27.5	
63	4/5	R	205.3	−1.8	157	62	182.8	−42.6	114.9	8.6	
64	5/9	R	194.9	39.5	141	63	202.7	−15.7	30.5	16.0	
65	6/7	N	358.5	−33.4	136	74	4.3	22.6	28.0	12.9	
Averages											
A	7		183.2	−6.0			–	–	2.9	43.6	
			–	–			11.6	33.0	19.9	14.0	6.86
C	4		15.7	−16.1			–	–	9.6	31.3	
			–	–			8.5	33.8	17.5	22.6	1.82

Column headings as in Table 2.

dates the Cenozoic folding and is inferred to be primary, although a remagnetization prior to folding always remains a probability.

5. Discussion

The data have also been plotted as poles, in Fig. 15a. The consistency between sub-studies is again obvious. The four poles for both the Lower and Upper Cretaceous from the Yingjisha and Wuqia areas lie within the joint intersection of their 95% confidence intervals and are, therefore, statistically indistinguishable.

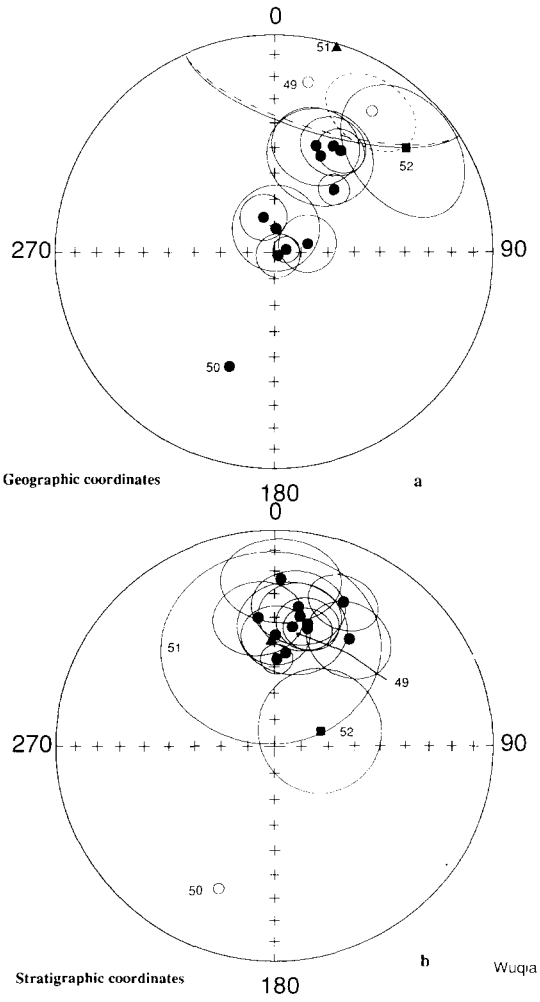


Fig. 12. As in Fig. 8 but for the Upper Cretaceous sites of Wuqia area. Square = outlier.

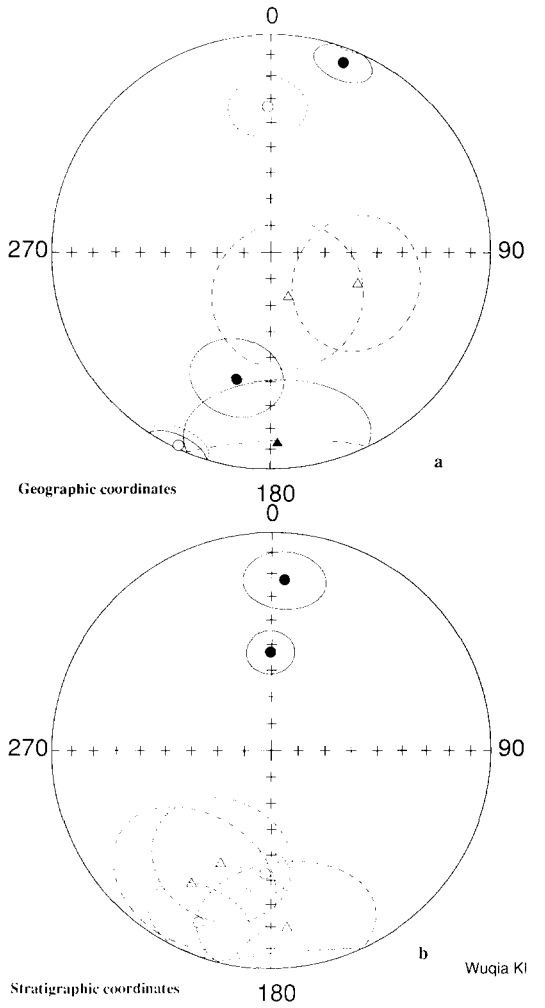


Fig. 13. As in Fig. 8 but for the Lower Cretaceous sites of the Wuqia area.

Data from roughly the same periods are available from northern Tarim [1] and southern Junggar [2] (Fig. 15b and Table 8). These data are respectively dated from about the Jurassic/Cretaceous and Cretaceous/Tertiary boundaries and can be compared respectively to our lower Cretaceous and upper Cretaceous data. The Junggar data are in remarkable agreement with the Wuqia and Yingjisha western Tarim data. They are both within the joint intersection of the six 95% confidence intervals (with a radius of roughly 4°)! The Upper Cretaceous data from northern Tarim are also very close, whereas the Lower Cretaceous data appear to be slightly displaced.

Close examination of the remarkable set of eight poles shown in Fig. 15b shows that the lower Cretaceous poles from the four areas are not statistically distinct from their upper Cretaceous counterparts (Table 8). This can be interpreted as meaning that there was no significant motion of each area with respect to a reference system tied with the Earth's rotation axis (but hopefully not that the stratigraphic distinction between Ku and Kl was illusory) and the Lower and Upper Cretaceous data can be averaged in each area. This results in the four mean poles shown in Fig. 16. The Cretaceous mean poles for the Wuqia, Yingjisha and Junggar are indistinguishable. In contrast, *the Uytak pole is significantly displaced from the Wuqia and Yingjisha poles*, and the northern Tarim pole is significantly displaced from the Junggar pole. This *last* observation was made by Chen et al. [2], when they compared their Junggar data with the Tarim data of Li et al. [1] and was taken as support for the tectonic model of Avouac et al. [28] for the formation of the Tien Shan mountains. However appealing, this conclusion relied on scarce data and the rather large uncertainty in the Li et al. data made the angular differences barely significant ($8.7 \pm 7.5°$). This intriguing result was one of the incentives for returning to Tarim and increasing the data base in number and geographical coverage. The new results presented in this paper lead to a somewhat revised interpretation.

The remarkable consistency of the Wuqia, Yingjisha and Upper Cretaceous northern Tarim poles (i.e., five studies) leads to the suggestion that the Tarim block formed a single tectonic unit prior to and during the Cretaceous and did not undergo subsequent tectonic deformation of sufficient extent to be measurable by paleomagnetism. Despite obvious deformation, these areas can apparently safely be tied to the Tarim craton. Since the other data are located on a small circle centred in Tarim, it appears much more reasonable to assume that the three slightly to significantly discordant poles are due to a local rotation about vertical axes.

As recalled in the section on geology, the Uytak sites were collected from an area with intense deformation, in particular NW–SE trending thrusts and right-lateral strike-slip faults (Figs. 1 and 2). This may have been responsible for the

TABLE 8

Cretaceous paleomagnetic results from Tarim and Junggar basins

Block	Area	Age	N	S Lat.	S Long.	P Lat.	P Long.	A95 (dp/dm)	Reference
Tarim	Yingjisha	Ku	6	38.5	76.4	71.0	234.0	6.8/11.6	This study
Tarim	Yingjisha	Kl	3	38.5	76.4	70.4	212.1	6.6/10.8	This study
Tarim	Yingjisha	K	9	38.5	75.5	71.2	226.7	4.7/8.0	This study
Tarim	Wuqia	Ku	11	39.5	75.0	70.8	222.6	5.4/8.9	This study
Tarim	Wuqia	Kl	7	39.5	75.0	66.3	226.6	9.0/15.9	This study
Tarim	Wuqia	K	18	39.5	75.0	70.1	225.8	4.9/8.2	
Tarim	Kuche	Ku	4	41.6	83.5	66.2	223.3	6.1/10.3	[1]
Tarim	Paicheng	K/J *	6	41.8	82.0	64.4	208.8	6.4/10.5	[1]
Tarim	N. Tarim	K	10	41.7	82.7	65.4	214.0	6.1	
Junggar	Manas	K/Ter	9	44.2	86.0	74.3	223.1	6.4/9.4	[2]
Junggar	Manas	J/K	13	44.2	86.0	72.2	227.2	4.8/7.2	[2]
Junggar	Manas	K	22	44.2	86.0	73.4	223.6	4.3	
Tarim	1 (a)	K	6	39.0	75.6	67.9	211.7	7.0	
Tarim	2 (b)	K	5	36.6	77.2	69.1	223.8	3.5	
NCB		K	3			83.2	208.0	9.3	[29]
SCB		K	7			77.9	206.4	5.2	[29]
Eurasia	65–130 Ma		29			76.3	202.6	2.2	[30]

(a) Average of the six Ku and Kl studies reported in this paper (Uytak, Yingjisha and Wuqia).
(b) Average of five Ku and Kl studies, excluding Uytak and Paicheng indicated by * (see text).
NCB, SCB = North China block and South China block.
K = the average of the Upper and Lower Cretaceous.

dispersal of declinations even after application of the classical tilt correction. It is quite likely that this correction (i.e., assuming horizontal cylindrical folds) does not adequately describe the integral history of tectonic phases and may have left out some amount of deformation that can be described by rotation about vertical axes. Such shearing may have led both to within- and between-site dispersions and resulted in a large clockwise dispersion of declinations.

The Paicheng sites of Li et al. [1] are particularly interesting in this respect. When only the two studies of Li et al., at Paicheng for Jurassic/Cretaceous and Kuche for Upper Cretaceous (Fig. 1) were available, the fact that both were just within the intersection of their confidence intervals led Li et al. [1] and Chen et al. [2] to average them. Now, with the increased data base, there is an indication that the Paicheng directions may, in fact, have been rotated by some $8.2 \pm 12.1°$ with respect to the other, extremely well clustered, Tarim directions. The coordinates of the Paicheng sites are erroneously given in Table 1 of Li et al., but the small cartoon in their figure 2 leaves no doubt as to the exact sampling location.

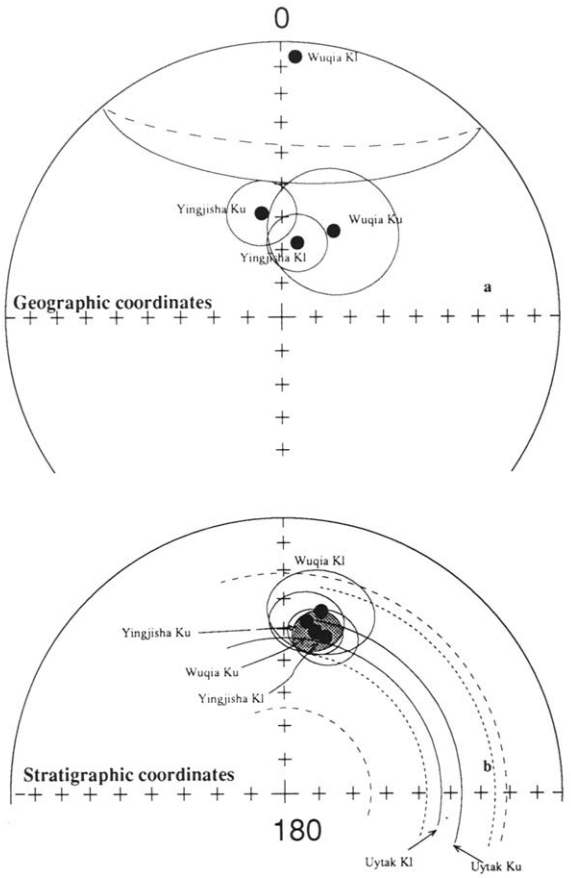


Fig. 14. Equal-area projection of mean directions from the three localities sampled (Yingjisha and Wuqia) and two age groups, showing that directions are much better grouped in stratigraphic coordinates. The circles show the inclination means (solid) and their error bars (dash).

Examination of the Landsat image shows that the sites were collected from the northeast trending limb of a characteristic re-entrant at the southern base of the Tien Shan range. This re-entrant corresponds to a rather steeply plunging, possibly conical, fold with a dipping (and not horizontal) axis. The traditional unfolding correction would, therefore, not be adequate. In order to apply a better correction, one needs to know not only the attitude of the beds that were collected (strike = 51.2°E, dip = 70.8°SE recalculated from the Li et al. data) but also the attitude of the beds immediately to the east of the fold hinge, which is unfortunately not available. Examination of the Landsat image and topography suggest that the strike may range from 90° to 110°E; the dip is apparently quite steep and is assumed to lie somewhere between 50° and 70°S. The bedding correction based on the plunge of the hinge axis is extremely sensitive to the unknown attitude of the eastern part of the fold; applying the extreme values listed above, the corrected declination lies anywhere between 0° and 50° (the value proposed by Li et al. [1] being 22°). Although it would be desirable to return to Paicheng to measure bedding attitudes accurately, it can be suggested that the paleomagnetic mean of Li et al. [1] has not been properly corrected for tectonic deformation.

We conclude that the present best estimate of a Cretaceous pole for the Tarim can be derived from the average of the Yingjisha and Wuqia Kl and Ku data and the Kuche Ku data (Table 8). This pole lies at 69.1°N, 223.8°E ($A_{95} = 3.5°$) and is shown in Fig. 17. The Cretaceous poles from Junggar [2], the North and South China blocks [29] and the apparent polar wander path for Eurasia from [30] are also shown. We have also calculated a mean pole from Eurasia by averaging all Cretaceous poles (Table 8). Notice that the

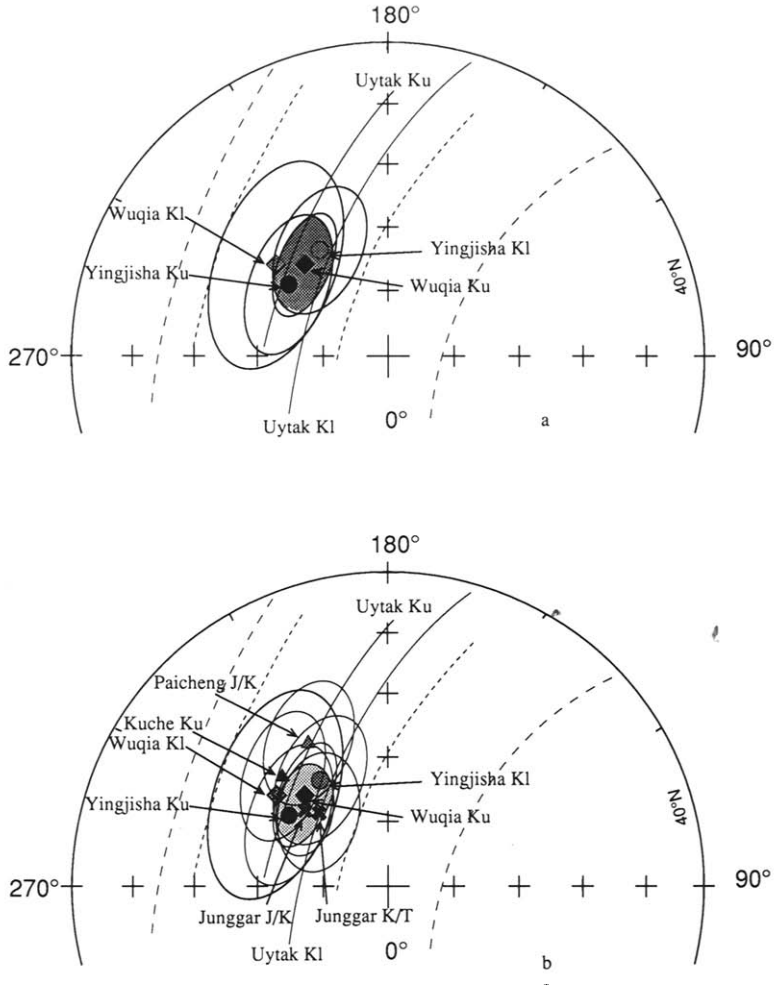


Fig. 15. Equal-area projection of mean poles (a) from this study and with those from North Tarim [1] and (b) South Junggar [2] added. The small circles centered on Uytak show the possible pole position on local rotation.

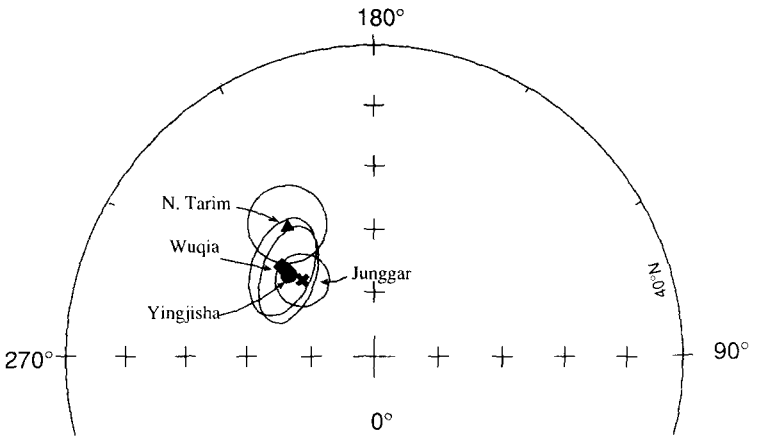


Fig. 16. Equal-area projection of Cretaceous mean poles (Lower and Upper Cretaceous averaged together) from the five areas sampled in Tarim and Junggar.

Tarim pole is based on five sub-studies spanning some 700 km, whereas the Junggar pole relies on two sub-studies along a single 200 km east–west traverse west of Urumqi.

The Tarim, Junggar and Eurasian poles lie along a great circle (paleomeridian) that transects the Tarim, Junggar and Eurasian blocks (or plates) in a north-northeast direction. They occur in a sequence that is consistent with the occurrence of shortening between each pair of blocks since the Cretaceous. Convergence between Junggar and Eurasia amounts to 640 ± 530 km, with no rotation (2.2° CCW $\pm$ 5.4°) as found by Chen et al. [2]. Convergence between Tarim and Junggar amounts to 400 ± 600 km, again with no rotation (2.6° CW $\pm$ 6.1°). Due to the reinterpretation of the Paicheng data, this is somewhat different from the values proposed by Chen et al. [2] (330 ± 760 km and 8.6 ± 8.7°, respectively). The order of magnitude of the shortening is the same and is consistent with the geology [28], but it is not statistically resolvable, and neither is the rotation. We insist, however, that the analysis proposed by Avouac et al. [28], based on independent geological observation and geophysical reasoning, remains valid and compatible with the paleomagnetic data. The obvious lack of rotation between the Tarim and Siberia could, of course, correspond to the addition of finite rotations of opposite sense cancelling each other, but we have no data at present to support this more complex solution.

The Tarim pole is significantly different from the NCB pole, which would imply that the two did not form a single undeformed block throughout the Mesozoic and Cenozoic. The polar difference leads to a latitudinal discrepancy (for a reference site at 40°N, 85°E) of 1400 ± 1060 km. Despite the rather large uncertainty, it seems that the Tarim and Junggar continued to penetrate northwards within Siberia in the Cenozoic (resulting in the rejuvenation of the Tien Shan and Sayan-Tuva ranges, respectively), at a time when the North and South China blocks were assembled with Eurasia and the Mongol–Okhotsk ocean was closed [29].

An alternate interpretation would be that the new loop in the APWP of Eurasia advocated by our group [30–33] might extend even further to the east than previously thought. A 10° extension in longitude would reduce the estimates of shortening between the Junggar, Tarim and Eurasia to unresolvable values of only a few hundred kilometres. On the other hand, the North China block and South China block data would be less concordant with Eurasia. Notice that a very sharp and elongated hairpin in an APWP is always a very difficult feature to resolve and that any pro-

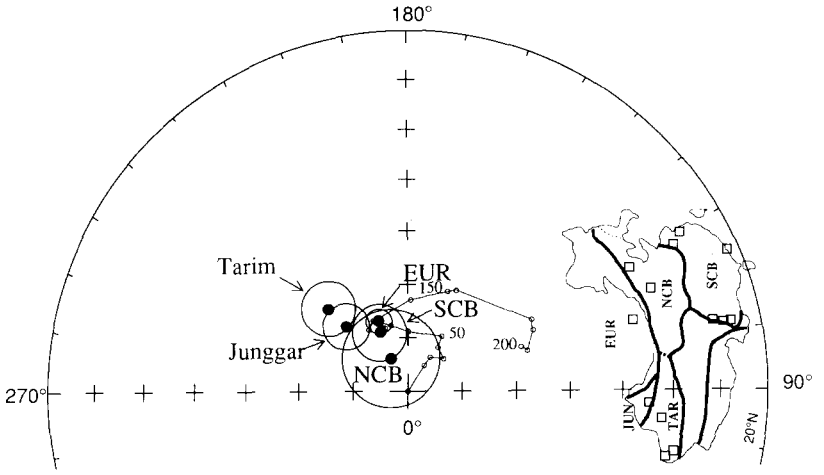


Fig. 17. Equal-area projection of Cretaceous mean poles from Tarim, Junggar, NCB, SCB and Eurasia (Table 8). The APWP of Eurasia from [30] for the last 200 Ma is also given (one point every 10 Ma). Sites from the various blocks of China which contribute data to the poles are shown on the map (see Table 8 for data).

cess that smooths the data suppresses it. A similar situation occurs with the Triassic hairpin in the APWP of North America [34,35].

Of course, it would be desirable to pursue work on the Cretaceous of North and South China to try to constrain their poles more precisely. However, the most urgent programe would be to re-sample extensively the Siberian plate to the North of the Junggar and North China blocks, since the Eurasian APWP we use as a reference [30] does not include any data from that plate.

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